

Deep Learning for Breast Cancer Classification using Ultrasound Images: Enhancing Diagnosis with Preprocessing and Model Optimization

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Outline



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Motivation



- Breast cancer remains one of the leading causes of cancer-related deaths among women worldwide. Early detection significantly improves patient outcomes, and breast ultrasound imaging is widely used as a safe, radiation-free, and cost-effective screening tool—especially for younger women or women with dense breast tissue.
- The [Breast Ultrasound Images Dataset \(BUSI\)](#) available on Kaggle provides a publicly accessible collection of normal, benign, and malignant ultrasound images. This dataset offers a valuable opportunity to explore and compare the performance of different deep learning models for breast lesion classification. For the purpose of this course project, it is well-suited in terms of size, clinical relevance, and ease of preprocessing.

Problem Description



- The BUSI dataset consists of breast ultrasound images labeled as **normal**, **benign**, or **malignant**. In this project, we need to classify the input images.
- Several challenges arise from this problem:
 - a. Class imbalance: The number of images across normal, benign, and malignant categories is uneven.
 - b. High image variability: Ultrasound images contain speckle noise, different tumor shapes, and variability in acquisition quality.
 - c. Limited dataset size: With relatively few samples, both simple models and deep networks face a risk of overfitting.

Goal



- The primary goal is to develop a robust framework for breast lesion classification by comparing:
 - a. **Baseline:** A Custom CNN with simple architecture (as a benchmark).
 - b. A transfer learning model based on **ResNet50**, initialized with ImageNet weights.
 - c. **Optimization Strategy:** Try to integrate the ResNet50 framework with:
 - i. Two-Stage Training to adapt feature extractors to medical textures.
 - ii. Speckle Denoising to improve image quality. (**Not used eventually**)
 - iii. Focal Loss to address class imbalance and maximize Recall.

Related Works

Key Reference

- Title: EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks[1]
- Problem/Challenge:
 - Arbitrary Scaling: Previous models scaled depth, width, or resolution arbitrarily, leading to suboptimal efficiency and accuracy.
 - Computation vs. Accuracy: Achieving high accuracy often requires massive parameters
- Approach :
 - Compound Scaling: Proposed a novel scaling method that uniformly scales network depth, width, and resolution using a compound coefficient.
 - Neural Architecture Search (NAS): Used NAS to design a lightweight baseline network
- Limitation :
 - High Risk of Overfitting on Small Data: The model is optimized for ImageNet (14M images). On small medical datasets, its complex feature extractors can easily overfit.
 - Sensitivity to Speckle Noise: EfficientNet assumes clean RGB inputs. It lacks explicit mechanisms to filter out ultrasound speckle noise, leading to potential misclassification.

Work Comparison



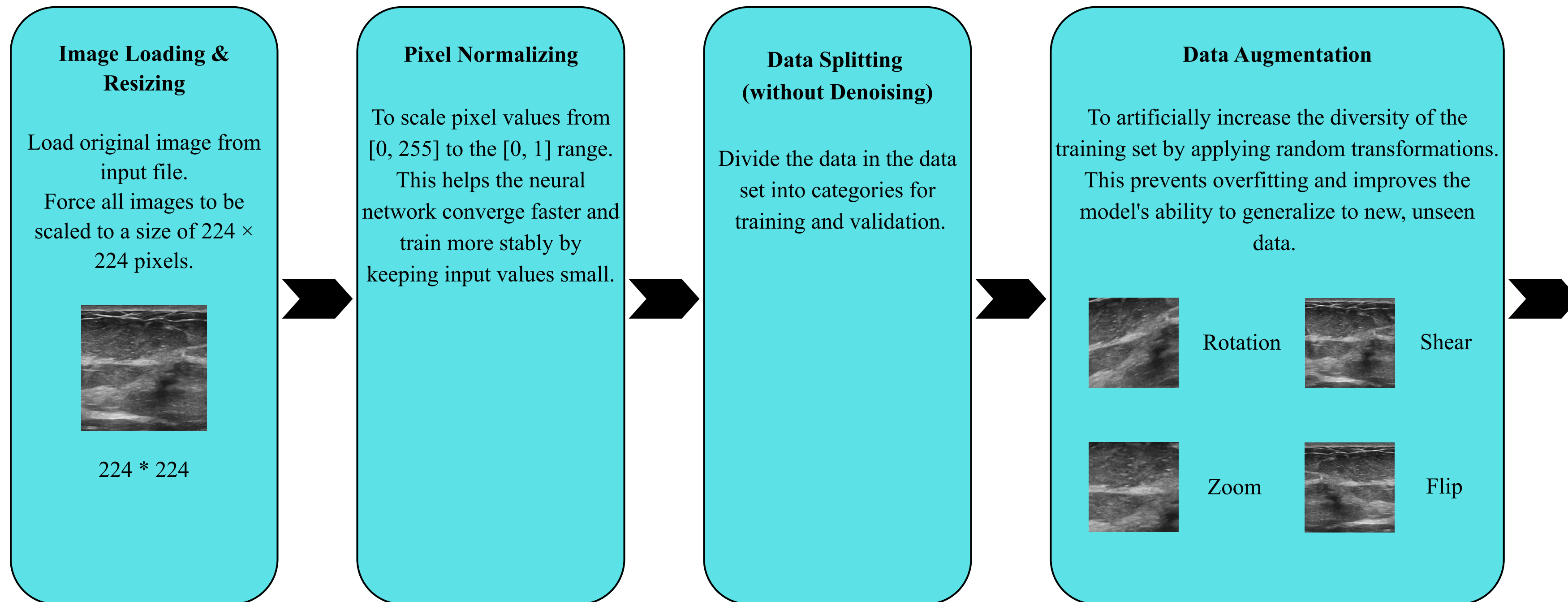
	Baseline CNN (e.g., VGG16[1])	EfficientNet[2]	Proposed Framework
Data Hunger	Medium	High	Low
Training Stability	High	Medium	High
Overfitting Risk	High	High	Low
Speckle Noise Handling			V
Class Imbalance Handling			V

[1] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," in International Conference on Learning Representations (ICLR), 2015.

[2] M. Tan and Q. V. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," in Proceedings of the 36th International Conference on Machine Learning (ICML), 2019, pp. 6105-6114.

Proposed Method

The Architecture of Proposed Model



The Architecture of Proposed Model



Build Model

1. ResNet50 Backbone (Transfer Learning)
2. Global Average Pooling
3. Classification Head (Dense + Dropout)
4. Compilation with Focal Loss
5. Callbacks & Two-Stage Training



Fitting Model

The model uses preprocessed data to learn patterns and update its weights.



Model Evaluation & Visualization

Calculates accuracy and loss on the validation set to quantify the model's effectiveness.
Then, plots the training history to visualize learning trends.

The Detail of Methods - Data Preprocessing

1. Image Loading & Resizing

- Load original images from the dataset.
- Resize all images to a consistent target resolution of 224×224 pixels to match the input requirement of the ResNet50 backbone.

2. Pixel Normalizing

- Scale pixel values from $[0, 255]$ to the $[0, 1]$ range.
- This helps the neural network converge faster and train more stably by keeping input gradients consistent.

3. Denoising

- We apply a Median Filter to the training and validation sets individually.
- This effectively removes ultrasound speckle noise while preserving tumor boundaries, preventing the model from learning noise artifacts as features.

4. Data Augmentation

- To artificially increase the diversity of the training set, we apply random transformations:
 - Rotation, Shearing, Zooming, and Horizontal Flipping.
- This prevents overfitting on our small dataset (780 images) and improves the model's ability to generalize to unseen data.

The Detail of Methods - Proposed Model

1. Model Construction (Backbone)

- We employed ResNet50 initialized with ImageNet weights as the feature extractor.
- The residual connections enable training deeper networks without gradient degradation, thereby capturing complex tumor features.

2. Two-Stage Training Strategy

- Phase 1 (Warm-up): We freeze the ResNet50 backbone and train only the custom classifier head to stabilize weights.
- Phase 2 (Fine-tuning): We unfreeze the top 50 layers and train with a very low learning rate. This adapts the high-level features specifically for ultrasound textures.

3. Loss Function Optimization

- Instead of standard Cross-Entropy, we implemented Focal Loss during compilation.
- This addresses the class imbalance by assigning higher weights to hard-to-classify examples.

4. Training Callbacks

- EarlyStopping: Prevents overfitting by monitoring validation loss.
- ModelCheckpoint: Saves the best model weights.
- ReduceLROnPlateau: Dynamically lowers the learning rate when training stagnates.

Experiment

Data Description

This study uses the Breast Ultrasound Images Dataset, which contains medical ultrasound images collected from female patients for breast cancer screening.

Participants: 600 women, ages 25–75

Year of Collection: 2018

Image Count: 780

Image Format: PNG

Average Resolution: 500×500 pixels

Classes:

- Normal
- Benign
- Malignant

Each image is paired with its ground truth label, enabling supervised learning tasks. This dataset is widely used for classification, detection, and segmentation of breast cancer using machine learning.

Evaluation Metrics



1. Primary Metric: Recall ($TP / (TP + FN)$)

In cancer screening, False Negatives are dangerous (missing a cancer patient).

High Recall is our top priority to ensure no malignant tumor is overlooked.

2. Secondary Metric: F1-Score

Since the BUSI dataset is imbalanced, accuracy can be misleading.

F1-Score provides a fairer assessment of model performance on minority classes.

3. Training Monitoring: Loss & Accuracy

Validation Loss: Used to monitor convergence and prevent overfitting (via Early Stopping).

Accuracy: Provides a general overview of correct predictions across all classes.

Experiment List



- Baseline Evaluation: Evaluate the performance of the Custom Simple CNN, VGG16 and EfficientNet.
- Proposed Model Evaluation: Evaluate the ResNet50 model with Transfer Learning.
- Comparison: Compare Accuracy, Recall, and Stability between Baseline and Proposed models.
- Ablation Study: Analyze the impact of specific modules (Denoising and Focal Loss) on performance.

Experiment Results



Model Architecture	F1-Score	Recall	Status
Simple CNN	0.66	0.62	Referenced
VGG16	0.67	0.65	Referenced
EfficientNet	0.83	0.85	Referenced
Proposed ResNet50	0.98	0.98	Ours (Optimized)

Ablation Study



Experiment	Model Strategy	Denoising	Loss Function	Why it helps?
M1	ResNet50	✗	Cross-Entropy	Baseline (Standard Transfer Learning)
M2	ResNet50	✗	Focal Loss	Adapts loss function
M3	ReNet50	✓	Focal Loss	Removes speckle noise interference

Ablation Study



Experiment	Accuracy	Recall	Precision	Macro F1
M1	0.98	0.99	0.98	0.98
M2	0.98	0.99	0.98	0.98
M3	0.38	0.34	0.34	0.33

Ablation Study



- In our ablation study, we observe that there is a negative effect of median filtering on breast cancer classification.
- While median filtering is widely used to suppress noise in medical images, our results indicate that it introduces a distribution shift when applied to pretrained CNN models.
- The filtered images oversmoothed lesion regions and removed subtle but critical cues, leading to a severe degradation in recall (particularly for malignant cases), thereby increasing false negatives and reducing clinical reliability.

Conclusion



- In conclusion, ResNet achieves robust and clinically meaningful performance on breast cancer classification.
- Contrary to expectation, incorporating median filtering degrades performance by removing discriminative texture features and causing a mismatch with pretrained representations.
- Median filtering may still be beneficial when training models from scratch or for segmentation tasks, but it is not suitable for pretrained classification models in our setting.
- Therefore, we adopt ResNet with focal loss as our final model.

- [1] W. Al-Dhabyani, M. Gomaa, H. Khaled, and A. Fahmy, "Dataset of breast ultrasound images," Data in Brief, vol. 28, p. 104863, 2020.
- [2] M. Tan and Q. V. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," in Proceedings of the 36th International Conference on Machine Learning (ICML), 2019, pp. 6105-6114.
- [3] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 770-778.
- [4] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," in International Conference on Learning Representations (ICLR), 2015.
- [5] T. Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal Loss for Dense Object Detection," in Proceedings of the IEEE International Conference on Computer Vision (ICCV), 2017, pp. 2980-2988.

Thank you